SUMS OF POWERS VIA BACKWARD FINITE DIFFERENCES AND NEWTON'S FORMULA

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Abstract. In this manuscript, we derive closed formulas for multifold sums of powers of integers by combining the backward Newton interpolation formula with hockey-stick identities for binomial coefficients. We further obtain representations of multifold sums of powers in terms of Stirling numbers of the second kind and Eulerian numbers. Finally, we provide Wolfram Mathematica programs for the efficient verification of the derived identities.

Contents

Date: June 29, 2026.

2010 *Mathematics Subject Classification.* 05A19, 05A10, 11B68, 11B73, 11B83.

Key words and phrases. Sums of powers, Newton's interpolation formula, Finite differences, Binomial coefficients, Faulhaber's formula, Bernoulli numbers, Bernoulli polynomials, Interpolation, Approximation, Discrete convolution, Combinatorics, Polynomial identities, Central factorial numbers, Stirling numbers, Eulerian numbers, Worpitzky identity, Pascal's triangle, OEIS.

DOI: https://doi.org/10.5281/zenodo.18118011

1. Auxiliary lemmas

In this section, allow us to define a few lemmas, definitions, and conventions that serve as foundation for main results of this manuscript. For multifold sums of powers, we use the notation provided by Donald Knuth in [?], because it is simple and beautiful

Definition 1.1 (Multifold sums of powers). *For non-negative integers r, n, m*

$$\begin{aligned}
\Sigma^0 n^m &= n^m, \\
\Sigma^1 n^m &= \Sigma^0 1^m + \Sigma^0 2^m + \cdots + \Sigma^0 n^m, \\
\Sigma^{r+1} n^m &= \Sigma^r 1^m + \Sigma^r 2^m + \cdots + \Sigma^r n^m.
\end{aligned}$$

We utilize the following notation for rising factorials

Definition 1.2 (Rising factorials). *For integers n, k*

$$n^{(k)} = \begin{cases} 0, & \text{if } k < 0, \\ 1, & \text{if } k = 0, \\ \prod_{j=0}^{k-1}(n+j), & \text{if } k > 0. \end{cases}$$

We encourage the audience to read J. F. Steffensen's work on the definition of generalized factorials [?].

Lemma 1.3 (Backward Newton's formula). *Let $f\colon \mathbb{Z} \to \mathbb{C}$ be a function, and let $t \in \mathbb{Z}$. Then, for all $n \in \mathbb{Z}$,*

$$f(n) = \sum_{k=0}^{\infty} \frac{(n-t)^{(k)}}{k!} \nabla^k f(t) = \sum_{k=0}^{\infty} \binom{n-t+k-1}{k} \nabla^k f(t),$$

where $\nabla^k f(t)$ is the k-th backward difference of f evaluated at t

$$\nabla^k t^m = \sum_{j=0}^{k} (-1)^j \tbinom{k}{j} (t-j)^m.$$

Proof. Originally revealed by Sir Isaac Newton in [?]. Expressed in modern mathematical notation by J. F. Steffensen in [?, section 2, eq. (19)]. By using identity $\frac{(n-t)^{(k)}}{k!} = \binom{n-t+k-1}{k}$, we get binomial form. $\square$

Lemma 1.4 (Backward Newton's formula for powers)**.** *For non-negative integers $n, m,$ and an arbitrary integer t*

$$n^m = \sum_{k=0}^{m} \binom{n-t+k-1}{k} \nabla^k t^m.$$

Proof. Special case of Lemma (1.3) for powers. □

Lemma 1.5 (Generalized hockey-stick identity)**.** *For arbitrary integers a, b and j,*

$$\sum_{k=a}^{b} \binom{k}{j} = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

Proof. We have, $\sum_{k=a}^{b} \binom{k}{j} = \binom{a}{j} + \binom{a+1}{j} + \cdots + \binom{b}{j} = \left(\sum_{k=0}^{b} \binom{k}{j}\right) - \left(\sum_{k=0}^{a-1} \binom{k}{j}\right)$. By hockey-stick identity $\sum_{k=0}^{n} \binom{k}{j} = \binom{n+1}{j+1}$ yields,

$$\sum_{k=a}^{b} \binom{k}{j} = \left(\sum_{k=0}^{b} \binom{k}{j}\right) - \left(\sum_{k=0}^{a-1} \binom{k}{j}\right) = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

This completes the proof. □

Lemma 1.6 (Backward hockey-stick identity)**.** *For integer $n \geq 1$, and arbitrary integers t, k, r*

$$\sum_{j=1}^{n} \binom{j-t+k-1+r}{k+r} = \binom{n-t+k+r}{k+r+1} - \binom{k-t+r}{k+r+1}.$$

Proof. By setting $a = k - t + r$, and $b = n - t + k - 1 + r$ into Lemma (1.5). □

Lemma 1.7 (Multifold sums of zero powers)**.** *For non-negative integers r, n*

$$\Sigma^r n^0 = \binom{r+n-1}{r}.$$

Proof.

(1) Let $r = 0$, then we have $\Sigma^0 n^0 = 1$, by definition (1.1).
(2) Let $r = 1$, then we have $\Sigma^1 n^0 = \sum_{k=1}^{n} 1 = \binom{n}{1}$.
(3) Let $r = 2$, then we have $\Sigma^2 n^0 = \sum_{k=1}^{n} \binom{k}{1} = \binom{n+1}{2}$.
(4) Let $r = 3$, then we have $\Sigma^3 n^0 = \sum_{k=1}^{n} \binom{k+1}{2} = \binom{n+2}{3}$.

(5) Similarly, for arbitrary integer r, by hockey-stick identity $\sum_{k=1}^{n}\binom{k}{r}=\binom{n+1}{r+1}$, yields $\Sigma^r\, n^0=\sum_{k=1}^{n}\Sigma^{r-1}\, k^0=\sum_{k=1}^{n}\binom{k+r-2}{r-1}=\binom{r+n-1}{r}$.

This completes the proof. □

Lemma 1.8 (Upper Binomial negation)**.** *For integers n, k*

$$\binom{-n}{k}=(-1)^k\binom{k+n-1}{k}$$

Proof. See [**?**, p. 164, eq. (5.14)]. □

Convention 1.9. *For all x*

$$x^0=1.$$

Donald Knuth give extensive discussion on the convention $x^0=1$ for all x, including zero, in Concrete Mathematics [**?**, p. 162], and in [**?**].

2. Introduction

In this manuscript, we derive formulas for sums of powers by combining backward Newton's interpolation formula for powers (1.4) with hockey-stick identity (1.6). We implement the algorithm for finding closed forms of sums of powers of integers, proposed in [**?**, Alg. (11.2)]. More precisely, the main idea is to determine binomial basis of Newton's formula, in terms of variants of difference operators, for example

$$\begin{cases} f(x) &= \sum_{k=0}^{\infty}\frac{(x-a)^{[k]}}{k!}\delta^k f(a)=\sum_{k=0}^{\infty}\frac{x-a}{k}\binom{x-a+\frac{k}{2}-1}{k-1}\delta^k f(a),\\ f(x) &= \sum_{k=0}^{\infty}\frac{(x-a)_k}{k!}\Delta^k f(a)=\sum_{k=0}^{\infty}\binom{x-a}{k}\Delta^k f(a),\\ f(x) &= \sum_{k=0}^{\infty}\frac{(x-a)^{(k)}}{k!}\nabla^k f(a)=\sum_{k=0}^{\infty}\binom{x-a+k-1}{k}\nabla^k f(a),\\ f(x) &= \sum_{k=0}^{\infty}\frac{(x-a)^k}{k!}\frac{\mathrm{d}^k f(a)}{\mathrm{d}x^k}.\end{cases}$$

We may observe that each variation of Newton's formula above has its binomial basis

$$\begin{cases} \frac{(x)_n}{n!} & = \frac{1}{n!}x(x-1)(x-2)\cdots(x-n+1) = \binom{x}{n}, \\ \frac{x^{(n)}}{n!} & = \frac{1}{n!}x(x+1)(x+2)\cdots(x+n-1) = \binom{x+n-1}{n}, \\ \frac{x^{[n]}}{n!} & = \frac{1}{n!}\left(n+\frac{k}{2}-1\right)\left(n+\frac{k}{2}-2\right)\cdots\left(n-\frac{k}{2}+1\right) = \frac{x}{n}\binom{x+\frac{n}{2}-1}{n-1}. \end{cases}$$

This allows us to successfully combine Newton's formula with hockey-stick type identity, to find closed forms for sums of powers of integers. Thus, we focus on closed forms of powers sums, by utilizing backward Newton's formula, and hockey-stick identity (1.6).

3. Main results

We start our discussion from Lemma (1.4), which yields backward Newton's interpolation formula for powers. For example,

$$\begin{aligned} n^3 &= 0\binom{n-1}{0} + 1\binom{n}{1} - 6\binom{n+1}{2} + 6\binom{n+2}{3}, \\ n^3 &= 1\binom{n-2}{0} + 1\binom{n-1}{1} + 0\binom{n}{2} + 6\binom{n+1}{3}, \\ n^3 &= 8\binom{n-3}{0} + 7\binom{n-2}{1} + 6\binom{n-1}{2} + 6\binom{n}{3}, \\ n^3 &= 27\binom{n-4}{0} + 19\binom{n-3}{1} + 12\binom{n-2}{2} + 6\binom{n-1}{3}. \end{aligned}$$

Thus, the transition to formula for sums of powers is quite straightforward,

$$\Sigma^1 n^m = \sum_{k=0}^{m}\left[\sum_{j=1}^{n}\binom{j-t+k-1}{k}\right]\nabla^k t^m.$$

Therefore, by setting $r=0$ into Lemma (1.6) yields closed form of ordinary sums of powers

Proposition 3.1 (Ordinary sums of powers)**.** *For non-negative integers n, m, and an arbitrary integer t*

$$\Sigma^1 n^m = \sum_{k=0}^{m}\left[\binom{n-t+k}{k+1} - \binom{k-t}{k+1}\right]\nabla^k t^m.$$

We may see that the basis $\binom{j-t+k-1+r}{k+r}$ for repeatedly applied Lemma (1.6) was indeed chosen correctly, because we derived closed form for sums of powers with ease, simply setting parameter for r. For example,

Example 3.2. *For integer $0 \leq t \leq 3$, we have*

$$\Sigma^1 n^3 = 0\left[\binom{n}{1} - \binom{0}{1}\right] + 1\left[\binom{n+1}{2} - \binom{1}{2}\right] - 6\left[\binom{n+2}{3} - \binom{2}{3}\right] + 6\left[\binom{n+3}{4} - \binom{3}{4}\right],$$

$$\Sigma^1 n^3 = 1\left[\binom{n-1}{1} - \binom{-1}{1}\right] + 1\left[\binom{n}{2} - \binom{0}{2}\right] + 0\left[\binom{n+1}{3} - \binom{1}{3}\right] + 6\left[\binom{n+2}{4} - \binom{2}{4}\right],$$

$$\Sigma^1 n^3 = 8\left[\binom{n-2}{1} - \binom{-2}{1}\right] + 7\left[\binom{n-1}{2} - \binom{-1}{2}\right] + 6\left[\binom{n}{3} - \binom{0}{3}\right] + 6\left[\binom{n+1}{4} - \binom{1}{4}\right],$$

$$\Sigma^1 n^3 = 27\left[\binom{n-3}{1} - \binom{-3}{1}\right] + 19\left[\binom{n-2}{2} - \binom{-2}{2}\right] + 12\left[\binom{n-1}{3} - \binom{-1}{3}\right] + 6\left[\binom{n}{4} - \binom{0}{4}\right].$$

From example above, we may observe binomial coefficients of negative upper index, which may looking unfamiliar. However, there is no reason to be afraid of, because negative upper index binomial coefficients are completely deterministic, because $\binom{-n}{k} = (-1)^k\binom{k+n-1}{k}$. Moreover, binomial coefficient $\binom{n}{k}$ is a polynomial in n, thus accepting even fraction values in its upper index. Therefore, Proposition (3.1) is also a polynomial in n. For example,

Example 3.3. *For integer $0 \leq t \leq 4$, we have*

$$t = 0: \quad \Sigma^1 n^3 = n \cdot 0 + \tfrac{1}{2}(n + n^2) \cdot 1 + \tfrac{1}{6}(2n + 3n^2 + n^3) \cdot (-6) + \tfrac{1}{24}(6n + 11n^2 + 6n^3 + n^4) \cdot 6,$$

$$t = 1: \quad \Sigma^1 n^3 = n \cdot 1 + \tfrac{1}{2}(-n + n^2) \cdot 1 + \tfrac{1}{6}(-n + n^3) \cdot 0 + \tfrac{1}{24}(-2n - n^2 + 2n^3 + n^4) \cdot 6,$$

$$t = 2: \quad \Sigma^1 n^3 = n \cdot 8 + \tfrac{1}{2}(-3n + n^2) \cdot 7 + \tfrac{1}{6}(2n - 3n^2 + n^3) \cdot 6 + \tfrac{1}{24}(2n - n^2 - 2n^3 + n^4) \cdot 6,$$

$$t = 3: \quad \Sigma^1 n^3 = n \cdot 27 + \tfrac{1}{2}(-5n + n^2) \cdot 19 + \tfrac{1}{6}(11n - 6n^2 + n^3) \cdot 12 + \tfrac{1}{24}(-6n + 11n^2 - 6n^3 + n^4) \cdot 6,$$

$$t = 4: \quad \Sigma^1 n^3 = n \cdot 64 + \tfrac{1}{2}(-7n + n^2) \cdot 37 + \tfrac{1}{6}(26n - 9n^2 + n^3) \cdot 18 + \tfrac{1}{24}(-50n + 35n^2 - 10n^3 + n^4) \cdot 6.$$

Where the coefficients

- For $t = 0$ are $0, 1, -6, 6$ belong to sequence A000000 in the OEIS.

- For $t = 1$ are $1, 1, 0, 6$ belong to sequence A000000 in the OEIS.
- For $t = 2$ are $8, 7, 6, 6$ belong to sequence A000000 in the OEIS.
- For $t = 3$ are $27, 19, 12, 6$ belong to sequence A000000 in the OEIS.
- For $t = 3$ are $64, 37, 18, 6$ belong to sequence A000000 in the OEIS.

Formula for double sums of powers could be derivate similarly, by setting $r = 1$ into Lemma (1.6) which yields

Proposition 3.4 (Double sums of powers)**.** *For non-negative integers n, m, and an arbitrary integer t*

$$\Sigma^2 n^m = \sum_{k=0}^{m} \left[\binom{n-t+k+1}{k+2} - \binom{k-t+1}{k+2} \Sigma^0 n^0 - \binom{k-t+0}{k+1} \Sigma^1 n^0 \right] \nabla^k t^m.$$

Note that we added correction terms $\Sigma^r n^0$, which are $\Sigma^0 n^0 = 1$, and $\Sigma^1 n^0 = n$. For example,

Example 3.5. *For integer $0 \le t \le 2$, we have*

$$\begin{aligned}
t = 0: \quad \Sigma^2 n^3 &= \tfrac{1}{2}(n + n^2) \cdot 0 + \tfrac{1}{6}(2n + 3n^2 + n^3) \cdot 1 + \tfrac{1}{24}(6n + 11n^2 + 6n^3 + n^4) \cdot (-6) \\
&\quad + \tfrac{1}{120}(24n + 50n^2 + 35n^3 + 10n^4 + n^5) \cdot 6, \\
t = 1: \quad \Sigma^2 n^3 &= \tfrac{1}{2}(n + n^2) \cdot 1 + \tfrac{1}{6}(-n + n^3) \cdot 1 + \tfrac{1}{24}(-2n - n^2 + 2n^3 + n^4) \cdot 0 \\
&\quad + \tfrac{1}{120}(-6n - 5n^2 + 5n^3 + 5n^4 + n^5) \cdot 6, \\
t = 2: \quad \Sigma^2 n^3 &= \tfrac{1}{2}(n + n^2) \cdot 8 + \tfrac{1}{6}(-4n - 3n^2 + n^3) \cdot 7 + \tfrac{1}{24}(2n - n^2 - 2n^3 + n^4) \cdot 6 \\
&\quad + \tfrac{1}{120}(4n - 5n^3 + n^5) \cdot 6.
\end{aligned}$$

Similarly, we obtain closed formula for multifold sums of powers, by using Lemma (1.6) repeatedly. Thus,

Theorem 3.6 (Multifold sums of powers)**.** *For non-negative integers r, n, m, and an arbitrary integer t*

$$\Sigma^r n^m = \sum_{k=0}^{m} \left[\binom{n-t+k+r-1}{k+r} - \sum_{s=0}^{r-1} \binom{k-t+r-1-s}{k+r-s} \Sigma^s n^0 \right] \nabla^k t^m.$$

For example,

Example 3.7. *For non-negative integers r, n, m we have*

$$\Sigma^r n^m = \sum_{k=0}^{m} \left[\binom{n+k+r-1}{k+r} - \sum_{s=0}^{r-1} \binom{k+r-1-s}{k+r-s} \Sigma^s n^0 \right] \nabla^k 0^m,$$

$$\Sigma^r n^m = \sum_{k=0}^{m} \left[\binom{n+k+r-2}{k+r} - \sum_{s=0}^{r-1} \binom{k+r-2-s}{k+r-s} \Sigma^s n^0 \right] \nabla^k 1^m,$$

$$\Sigma^r n^m = \sum_{k=0}^{m} \left[\binom{n+k+r-3}{k+r} - \sum_{s=0}^{r-1} \binom{k+r-3-s}{k+r-s} \Sigma^s n^0 \right] \nabla^k 2^m,$$

$$\Sigma^r n^m = \sum_{k=0}^{m} \left[\binom{n+k+r-4}{k+r} - \sum_{s=0}^{r-1} \binom{k+r-4-s}{k+r-s} \Sigma^s n^0 \right] \nabla^k 3^m.$$

By using Lemma (1.7), we get pure binomial version of Theorem (3.6)

Proposition 3.8 (Multifold sums of powers)**.** *For non-negative integers r, n, m, and an arbitrary integer t*

$$\Sigma^r n^m = \sum_{k=0}^{m} \left[\binom{n-t+k+r-1}{k+r} - \sum_{s=0}^{r-1} \binom{k-t+r-1-s}{k+r-s} \binom{s+n-1}{s} \right] \nabla^k t^m.$$

4. Stirling form

5. Eulerian form

6. Conclusions

In this manuscript, we derive closed formulas for multifold sums of powers of integers by combining the backward Newton interpolation formula with hockey-stick identities for binomial coefficients. We further obtain representations of multifold sums of powers in terms of Stirling numbers of the second kind and Eulerian numbers. Finally, we provide Wolfram Mathematica programs for the efficient verification of the derived identities.

Mathematica programs

Use this `GitHub` repository to validate the results using Mathematica programs.

Mathematica Function	**Validates / Prints**
`ValidateBackwardNewtonsFormulaForPowers.txt`	Validates Lem. (1.4)
`ValidateOrdinarySumsOfPowers.txt`	Validates Prop. (3.1)
`ValidateDoubleSumsOfPowers.txt`	Validates Prop. (3.4)
`ValidateMultifoldSumsOfPowers.txt`	Validates Thm. (**??**)

Metadata

- **Initial release date:** January 1, 2026.
- **Current release date:** June 29, 2026.
- **Version:** `Local-0.1.0`
- **MSC2010:** 05A19, 05A10, 11B68, 11B73, 11B83 .
- **Keywords:** Sums of powers, Newton's interpolation formula, Finite differences, Binomial coefficients, Faulhaber's formula, Bernoulli numbers, Bernoulli polynomials, Interpolation, Approximation, Discrete convolution, Combinatorics, Polynomial identities, Central factorial numbers, Stirling numbers, Eulerian numbers, Worpitzky identity, Pascal's triangle, OEIS .

- **DOI:** `https://doi.org/10.5281/zenodo.18118011`
- **Web Version:** `kolosovpetro.github.io/sums-of-powers-backward-differences`
- **Sources:** `github/kolosovpetro/SumsOfPowersBackwardDiffNewtonFormula`

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